

BEECH KING AIR 350i

Beginner Guide – Checklist & Procedures for MS Flight Simulator

by JayDee v0.3



This checklist is intended for flight simulator use only. If necessary I tried to optimize workflow for single pilot operation. Bold items are important & required, but sufficient for a quick startup. Non-bold items are optional for the complete checklist.

Diese Checkliste ist nur für den Gebrauch im Simulator gedacht. Wo es angebracht war, habe ich die Checkliste die Operation mit einem einzelnen Piloten optimiert. Fett gedruckte Punkte sind wichtig & notwendig, aber ausreichend für einen schnellen Startup. Nicht fett gedruckte Punkte sind optional für eine komplette Checkliste.

SPECIFICATIONS / LIMITATIONS

Max. Takeoff Weight.....	15.000 lbs
Max. Landing Weight.....	15.000 lbs
Service Ceiling.....	35.000 ft
Max. Demonstrated Crosswind (manual).....	20 kts
Max. Demonstrated Crosswind (AP coupled).....	34 kts
V _{NO} (Max. Structural Cruising Speed).....	263 KIAS
M _{MO} (Max. Structural Cruising Speed).....	M 0.58
V _A (Max. Maneuver Speed).....	184 KIAS
V _{FE TO} (Max. Flaps Extended Speed).....	202 KIAS
V _{FE FULL} (Max. Flaps Extended Speed).....	158 KIAS
V _{LE/LO} (Landing Gear Extension).....	184 KIAS
V _{SO} (Stall Speed Full Flaps).....	81 KIAS
Range.....	~ 1.500 NM
Takeoff Ground Roll (Sea Level).....	~ 2.500 – 3.500 ft
Landing Distance (Sea Level).....	~ 3.000 – 3.500 ft

FUEL PLANNING

For a Max. Power Flight load roughly 0.45 Gallons per NM.
For a 45% Power Flight load roughly 0.35 Gallons per NM

Trip Fuel in Gallons (very roughly and with reserve)					
Dist. in NM	300	600	900	1200	1500
@MaxPower	135 gal	270 gal	405 gal	540 gal	-----
@45%Power	105 gal	210 gal	315 gal	420 gal	525 gal

PREFLIGHT

FUEL/PAYLOAD Menu.....	SET & CHECK CG
Aircraft Documents.....	CHECKED
Parking Brake.....	SET
Landing Gear Control.....	DOWN
Elevator Trim.....	2 UNITS NOSE UP
Rudder and Aileron Trim.....	SET NEUTRAL
Battery Bus (MP).....	NORM
Voltmeter (OH).....	CHECK
a) Battery Position.....	MIN 23 VOLTS
b) Voltmeter All Other Postions.....	0 VOLTS
Battery (MP).....	ON
Landing Gear.....	ILLUMINATED
Fuel Quantity.....	CHECK MAIN & AUXILIARY
Voltmeter (OH).....	CHECK
a) TRIPPLE FEED.....	MIN 22 VOLTS
b) R and L GEN.....	0 VOLTS
c) CENTER.....	MIN 23 VOLTS
d) EXT PWR.....	0 VOLTS

Flaps.....	CHECK
a) Select APP and check if stoped at TO AND APP	
b) Select DOWN and check if stoped at DOWN	
c) Flaps UP	
Battery (MP).....	OFF
Oxygen System Preflight Check.....	COMPLETE

OUTSIDE CHECKS.....	COMPLETE
ATC Clearance.....	AS REQ
Pushback.....	AS REQ

ENGINE START (COLD)

Doors.....	LOCKED
Load and Baggage.....	SECURE
Passenger Briefing.....	COMPLETE
Cabin Equipment.....	POSITIONED
a) All Seats.....	SEAT BACKS UPRIGHT
b) Lateral-Tracking Seats.....	OUTBOARD POSITION
c) Tables.....	STOWED
Pilot Seats, Belts & Harness.....	ADJUSTED/FASTENED

Parking Brake.....	SET
ELT Switch.....	ARM
	Yellow XMT Light – EXTINGUISHED

Fuel Panel.....	CHECK
a) Standby Pumps.....	OFF
b) Crossfeed.....	OFF
c) Aux Transfer Switches.....	AUTO
d) Circuit Breakers.....	CHECK IN
Pilot's Instrument Panel and Subpanel.....	CHECK
a) STBY Flight Display System Power.....	TEST
b) STBY Flight Display System Power.....	ON
c) Battery Bus.....	CONFIRM NORM
d) Engine Anti-Ice.....	ON
e) Landing Gear Control.....	DOWN

EXT Power (MP).....	AS REQ
Battery (MP).....	ON
Avionics (MP).....	ON
Power Levers.....	IDLE
Propeller Levers.....	FULL FORWARD
Condition Levers.....	FUEL CUTOFF
Right Panel Circuit Breakers.....	CHECK IN
Overhead Switch Panel	
a) Master Panel Lights.....	AS REQ
b) Display Brightness.....	AS REQ
c) Interior Master Switch.....	ON

Beacon (MP).....	ON
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Right Ignition and Engine Start (MP).....	ON
Right Condition Lever (at min. 12 % N1).....	LOW IDLE
Right ITT and N1.....	MONITOR (max. 1000°)
(if no ITT rise is observed within 10 seconds, move the Condition Lever to CUTOFF and release Starter Switch)	
Right Oil Pressure.....	CHECK
Right Ignition and Engine Start (at min 50% N1).....	OFF
Right Condition Lever.....	HIGH IDLE
Right Generator (MP).....	ON

Repeat For Left Engine

Throttle Levers.....	min. 1.050 RPM
Voltmeter.....	CHECK
a) R and L GEN.....	27-29 VOLTS
Ext. Power (MP).....	OFF

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BEFORE TAXI

Avionics/GPS/Autopilot.....**SET & PROGRAMMED**
NAVI/CDI.....**SET GPS IF REQ**
Transponder.....**SET & STBY**
Altimeter.....**SET LOCAL QNH**

Interior Master Switch (OH).....AS REQ
Master Panel and Cockpit Lights (OH).....AS REQ
Cabin Lights (OH).....AS REQ
Cabin Signs(FSB Fasten Seats Belts)(OH).....ON
Standby Flight Display.....**INITIALIZATION COMPLETE**
Standby Altimeter.....**SET**
Prop Sync (MP).....ON
Flaps.....**UP**
Flight & Engine Instruments.....**CHECK**
Flight Controls.....**CHECK**

ATC Clearances.....AS REQ
Nav Lights (MP).....ON
Logo Lights (MP).....ON
Taxi Lights (MP).....ON
Parking Brake.....**RELEASE**
Brakes.....**CHECK**
Steering.....**CHECK**

BEFORE TAKEOFF

Avionics/GPS/Autopilot.....**CHECK SET**
NAVI/CDI.....**CHECK SET GPS IF REQ**
Transponder.....ON
Altimeter.....**CHECK SET LOCAL QNH**
Cabin Pressurization (CP).....1.000 ft ABOVE CRZ ALT
Pitch Trim.....**TAKEOFF**
Flaps.....**APPROACH**
Autofeather.....ARM
Propeller Levers.....MANUAL FEATHER CHECK
Fuel Quantity.....CHECK
Flight Instruments.....CHECK
Engine Instruments.....CHECK
V1, VR & V2.....SET
Engine Auto-Ignition (MP).....**ARM**
Anti-Ice (MP)
a) Engine Anti-Ice.....AS REQ
b) Windshield Anti-Ice.....**NORMAL/HI (if req.)**
c) Prop De-Ice.....**AUTO**
d) Left and Right Fuel Vent Heat.....ON
e) Stall Warning Heat.....ON
f) Pitot Heat L & R.....ON
Propeller Levers.....**FULL FORWARD**
Landing Lights (MP).....ON
Strobe Lights (MP).....ON
CAS Messages.....REMOVED OR ADRESSED

ATC Clearances.....AS REQ
Start Replay-Tool.....IF REQ
Parking Brake.....**RELEASE**

TAKEOFF (w. Flaps APPROACH)

Takeoff Power Limitations

Limits:.....100% TQ / 820° ITT / 104% N1 / 1.700 RPM

- line up and brake
- set Throttle to TAKEOFF POWER
- relase Brakes
- V1 ~ 95 - 100 KIAS (dep.on weight)
- rotate @ VR 102 - 105 KIAS (dep.on weight)
- Pitch ~ 10 - 15 °
- Initial Climb @ V2 105 - 110 KIAS (dep.on weight)
- retract GEAR

→ at ~ 400 ft AGL start to accelerate

VX (Best-Angle-Of-Climb).....125 KIAS

VY (Best-Rate-Of-Climb).....140 KIAS

VYSE (One Engine Inoperatvie -Best-Angle-Of-Climb)....125 KIAS

→ retract Flaps UP at 125 KIAS

CLIMB

Taxi Lights.....OFF
Flaps.....CHECK UP
Landing Gear.....CHECK RETRACTED
Yaw Damper.....ON (latest at 5.000 ft)
Windshield Anti-Ice.....CHECK NORMAL
Anti-Ice.....AS REQ
Cabin Pressurization.....CHECK

Cruise Climb Power Setting & Limitations

Limits:.....100% TQ / 785° ITT / 104% N1 / 1.700 RPM

Recommend Propeller Speed.....1.600 RPM

Cruise Climb Speed

SL to 10.000 ft.....170 KIAS

to 15.000 ft.....160 KIAS

to 20.000 ft.....150 KIAS

to 25.000 ft.....140 KIAS

to 30.000 ft.....130 KIAS

to 35.000 ft.....120 KIAS

Autopilot.AS REQ (min.1.000 ft AGL)

@ Transition Altitude

Altimeter.....**SET STD QNE**

@10.000 ft / FL100

Landing Lights.....**OFF**

CRUISE

Autofeather.....OFF
Cabin Pressurization.....SET AND CHECK
Anti-Ice.....AS REQ
Engine Auto-Ignition.....AS REQ. (ON IN ICING COND.)
Cabin Signs (FSB).....AS REQ

Cruise Climb Power Setting & Limitations

Limits:.....TQ see below/ 820° ITT / 104% N1 / 1.700 RPM

Recommend Propeller.....1.500 RPM

Maximum Cruise Power at ISA

SL to 20.000 ft.....100% TQ → ~ 260 – 240 KIAS

@ 22.000 ft.....95% TQ → ~ 225 KIAS

@ 24.000 ft.....85% TQ → ~ 217 KIAS

@ 26.000 ft.....80% TQ → ~ 209 KIAS

@ 28.000 ft.....75% TQ → ~ 200 KIAS

@ 30.000 ft.....70% TQ → ~ 190 KIAS

@ 33.000 ft.....60% TQ → ~ 175 KIAS

@ 35.000 ft.....55% TQ → ~ 162 KIAS

Normal Cruise Power at ISA

subtract 5 % TQ from maximum Cruise Power from 18.000 ft
on and above

Maximum Range Power

45 % TQ

DETERMINATION OF THE TOP OF DESCENT (TOD)

-----FOR IFR AND COMMERCIAL FLIGHTS MAINLY-----

Check STAR and APPROACH altitude constraints and if possible compare to actual charts. Estimate your TOD and compare it to actual TOD in FMS. Descent should start latest at approx. $(Altitude/1.000) \times 3$ NM +10 NM before the Destination. If ATC has not given you clearance 1-2 Minutes after your TOD, then start to descent anyways. Due to a lot of wrong STAR/APPROACH altitude constraints in MS FS database, you are advised to constantly check your altitude during descent and approach, even if you were given clearance to descent at the TOD. For a standard 3° glidepath approach, make sure you are not significantly higher at a given point than the elevation of the destination plus a 1.000 ft per 3 NM to go. Example: If you have 21 NM to go, destination is at 1.000 ft, you should be around at $21/3 \times 1.000 + 1.000$ ft = 8.000 ft altitude. Sinkrate for 3° Glidepath equals 5 * Ground Speed!

-- HAUPTSÄCHLICH FÜR IFR & KOMMERZIELLE FLÜGE--

Checkt die STAR und APPROACH Höhenbeschränkungen und vergleicht sie wenn möglich mit echten Karten. Überschlagt Euren TOD und vergleicht ihn mit dem TOD im FMS. Der Sinkflug sollte spätestens ca. bei $(Höhe/1.000) \times 3$ NM + 10 NM vor dem Ziel beginnen. Wenn Euch die ATC nicht 1-2 Minuten nach dem TOD die Freigabe zum Sinken erteilt hat, leitet den Sinkflug trotzdem ein. Auf Grund einer Vielzahl von falsch hinterlegten Höhenbeschränkungen in der Datenbak des MS FS, solltet Ihr regelmäßig eure Höhe überprüfen, auch wenn Ihr rechtzeitig die Sinkflugfreigabe erhalten hattet. Für einen Standard 3° Sinkpfad solltet Ihr sicherstellen, dass Ihr zu keinem Punkt höher als die Zielhöhe plus 1.000 ft pro 3 NM verbleibender Flug. Beispiel: Müsst ihr noch 21 NM fliegen und das Ziel liegt auf 1.000 ft Höhe, solltet Ihr bei ca. $21/3 \times 1000$ ft + 1.000 ft = 8.000 ft Höhe sein. Sie Sinkrate für einen 3° Sinkpfad entspricht dem 5 fachen der Ground Speed (Geschwindigkeit über Grund).

DESCENT

Avionics/GPS/Autopilot.....**SET**
 ILS Frequency.....**COMPARE AND CHECK SET**
 NAVI/Flightplan.....**ACTIVATE APPROACH IF REQUIRED**

Approach Speeds.....**CONFIRM**
 Cabin Pressurization.....**CHECK**
 Cabin Signs (FSB).....**AS REQ**
 Anti-Ice.....**AS REQ**
 Fuel Balance.....**CHECK**

Propeller Levers.....**1.500 RPM**
 Descent Speed.....**M.58 or 250 KIAS (wichever is less)**
 Descent Power.....**Maintain ~ 1.500 fpm**

Sinkrate for a 3° glidepath..... ~ 5 x Ground Speed
 (only as Information, not mandatory)

@ 10.000 ft / FL100

Landing Lights.....**ON/AS REQ**

@ Transition Flightlevel

Altimeter.....**SET LOCAL QNH**

APPROACH

@ ~ 15 NM to go

ATC Clearances.....**AS REQ**
 NAVI/CDI.....**SET VLOC IF REQ. FOR ILS**

Autofeather.....**ARM**
 Cabin Signs (FSB)(OH).....**ON**
 Cabin Pressurization.....**CHECK**

Altimeter Altimeter.....**CHECK SET LOCAL QNH**

VISUAL APPROACH & LANDING

Approach to enter Downwind Leg @ 1.000 ft AGL!

@ 8 - 7 NM to go

- reduce SPEED to ~ 150 - 160 KIAS
- set FLAPS 1 (APPROACH) @ ~ 160 KIAS

@ Downwind Leg or ~ 6 NM (@1.000 ft AGL)

- start reducing speed to ~ 120 KIAS
- lower LANDING GEAR (mid Downwind Leg)
- set Flaps 2 (DOWN) @ ~ 140 KIAS

@ Base Leg or ~ 3 NM (@1.000 ft AGL)

- YAW Damper OFF
- Propeller Levers FULL FORWARD
- start descending
- sinkrate ~ 500 – 1.000 fpm
- Airspeed on final VREF ~ 100 - 110 KIAS (dep.on weight)

@ Short Final

Flaps.....**CHECK FULL**
 Landing Gear.....**CHECK DOWN**
 Propeller Levers.....**CHECK FULL FORWARD**

@ Runway Threshold (@ ~ 50 ft)

- Throttle slowly to IDLE
- slowly flare @ 10 – 20 ft AGL
- touchdown with mainwheels first
- use Reverse Throttle if required

ILS APPROACH & LANDING

Approach to be @ ~ 3.000 ft above destination airport elevation or use proper procedure altitudes!

@ 15 – 12 NM to go

- reduce SPEED to ~ 150 - 160 KIAS
- set FLAPS 1 (APPROACH) @ ~ 160 KIAS

@ Glideslope Capture

- start descending
- sinkrate for 3° Glidepath = 5 x Ground Speed

@ ~ 6 NM to go

- start reducing speed to ~ 120 KIAS
- lower LANDING GEAR
- set Flaps 2 (DOWN) @ ~ 140 KIAS

@ ~ 3 NM to go

- Autopilot OFF
- YAW Damper OFF
- Propeller Levers FULL FORWARD
- Airspeed on final VREF ~ 100 - 110 KIAS (dep.on weight)

@ Short Final

Flaps.....**CHECK FULL**
 Landing Gear.....**CHECK DOWN**
 Propeller Levers.....**CHECK FULL FORWARD**

@ Runway Threshold (@ ~ 50 ft)

- Throttle slowly to IDLE
- slowly flare @ 10 – 20 ft AGL
- touchdown with mainwheels first
- use Reverse Throttle if required

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GO AROUND

- Throttle max. allowable Power (100% TQ)
- maintain VREF
- accelerate when clear of obstacles
- retract FLAPS to T.O. @ VREF +10
- retract LANDING GEAR
- Initial Climb @ ~ 125 KIAS
- Flaps UP at 200 ft AGL & min. 125 KIAS

AFTER LANDING

Taxi Lights.....	OFF
Landing Lights.....	OFF
Strobe Lights.....	OFF
Engine Auto-Ignition.....	OFF
Engine Anti-Ice.....	ON
Trasponder.....	STBY
Flaps.....	RETRACT
Cabin Pressurization.....	CHECK 0

PARKING / SHUT DOWN

Parking Brake.....	SET
Avionics.....	OFF
Autofeather.....	OFF
Exterior Lights.....	OFF
Interior Master Switch.....	OFF
ITT.....	STABILIZED
Condition Levers.....	FUEL CUTOFF
Propeller Levers.....	FEATHER
Overhead Panel Switches.....	OFF
Battery and Generators.....	OFF below 15% N1

ONE ENGINE OUT - FIRE OR FAILURE IN FLIGHT

OEI= One Engine Inoperative
OEI Best-Angle-of-Climb (VXSE)....125 KIAS
OEI Best-Rate-of-Climb (VYSE)....125 KIAS
OEI Enroute Climb.....125 KIAS
Air Minimum Control Speeds (VMCA)
 Flaps Up.....94 KIAS
 Flaps Approach.....93 KIAS
Emergency Descent.....184 KIAS
Maximum Glide Range.....135 KIAS

Bank the aircraft 3-5° into the operating engine while maintaining a constant heading.

- Condition Lever.....Fuel CutOFF
- Prop Lever.....FEATHER
- if Firewarning: Firewall Fuel Valve.....CLOSE
- if Firewarning persists: Fire Extinguisher...ACTUATE
- Generator.....OFF
- Engine Auto Ignition.....OFF
- Autofeather.....OFF